

NICHOLAS WILLIAMS

<https://nchlswllms.github.io> ♦ nw480@cam.ac.uk

PERSONAL INFORMATION

Full name: Nicholas James Williams
Date of birth: 11th July 1994
Place of birth: London, UK
Citizenship: UK

EDUCATION

Doctorate in Mathematics	<i>September 2018 – April 2022</i>
University of Cologne	<i>April 2021 – April 2022</i>
Started at University of Leicester	<i>September 2018 – April 2021</i>
Supervisor: Sibylle Schroll	
<i>Summa cum laude</i>	

Graduate Diploma in Quantitative Economics	<i>October 2017 – June 2018</i>
Birkbeck, University of London	
Distinction	

MMathPhil in Mathematics & Philosophy	<i>October 2012 – June 2016</i>
Worcester College, Oxford	
Double First Class Honours	

EMPLOYMENT

Research Associate	<i>October 2024 – September 2026</i>
University of Cambridge	
With Ailsa Keating	

Postdoctoral Research Associate	<i>September 2022 – September 2024</i>
Lancaster University	
With David Pauksztello	

JSPS Short Term Postdoctoral Research Fellow	<i>May 2022 – August 2022</i>
University of Tokyo	
With Osamu Iyama	

Graduate Teaching Assistant	<i>September 2021 – December 2021</i>
King's College London	

Guest Teacher	<i>September 2021 – December 2021</i>
London School of Economics	

Feedback Session Leader	<i>January 2019 – December 2020</i>
University of Leicester	

JOURNAL PUBLICATIONS

- | | |
|--|-------------|
| 10. A geometric perspective on the τ -cluster morphism category
With Sibylle Schroll, Aran Tattar, and Hipolito Treffinger
<i>Mathematische Zeitschrift</i> , Volume 312, article number 77 | <i>2026</i> |
| 9. The two higher Stasheff–Tamari orders are equal
<i>Journal of the European Mathematical Society</i> , Volume 28, no. 4, pp. 1391–1442 | <i>2026</i> |

8. Steenrod operations via higher Bruhat orders 2025
 With Guillaume Laplante-Anfossi
Proceedings of the London Mathematical Society, Volume 130, no. 2, e70024
7. A survey of congruences and quotients of partially ordered sets 2024
EMS Surveys in Mathematical Sciences, Volume 11, no. 1, pp. 153–203
6. Triangulations of prisms and preprojective algebras of type A 2024
 With Osamu Iyama
International Mathematics Research Notices, rnae059
5. Stability spaces of string and band modules 2024
 With Sibylle Schroll, Aran Tattar, Hipolito Treffinger, and Yadira Valdivieso
Journal of Pure and Applied Algebra, Volume 228, no. 4, 107503
4. Quiver combinatorics and triangulations of cyclic polytopes 2023
Algebraic Combinatorics, Volume 6, no. 3, pp. 639–60
3. The first higher Stasheff–Tamari orders are quotients of the higher Bruhat orders 2023
Electronic Journal of Combinatorics, Volume 30, no. 1, P1.29
2. New interpretations of the higher Stasheff–Tamari orders 2022
Advances in Mathematics, Volume 407, 108552
1. The combinatorics of tensor products of higher Auslander algebras of type A 2021
 With Jordan McMahon
Glasgow Mathematical Journal, 63(3), pp. 526–546

PEER-REVIEWED CONFERENCE PROCEEDINGS

4. From higher Bruhat orders to Steenrod cup- i coproducts 2024
 With Guillaume Laplante-Anfossi
Proceedings of the 36th International Conference on Formal Power Series and Algebraic Combinatorics, Séminaire Lotharingien de Combinatoire 91B, Art. 10
3. The higher Stasheff–Tamari orders in representation theory 2023
Representations of Algebras and Related Structures
2. Studying triangulations of even-dimensional cyclic polytopes via directed graphs 2023
Proceedings of the 35th International Conference on Formal Power Series and Algebraic Combinatorics, Séminaire Lotharingien de Combinatoire 89B, Art. 4
1. The first higher Stasheff–Tamari orders are quotients of the higher Bruhat orders 2021
Proceedings of the 33rd International Conference on Formal Power Series and Algebraic Combinatorics, Séminaire Lotharingien de Combinatoire 85B, Art. 6

OTHER CONFERENCE PROCEEDINGS

2. Donaldson–Thomas invariants for the Bridgeland–Smith correspondence 2024
Cluster Algebras and Its Applications
Oberwolfach Reports, No. 2/2024
1. Triangulations of cyclic polytopes and the higher Auslander algebras of type A 2022
Various issues on representation theory and related topics
RIMS Kōkyūroku, No. 2234

OTHER PUBLICATIONS

2. Microthesis: Triangulations of cyclic polytopes in representation theory 2022
London Mathematical Society Newsletter, No. 501, 33–34
1. Higher-dimensional combinatorics in representation theory 2022
 Doctoral thesis, University of Cologne

PREPRINTS

5. Relations between categorifications of higher-dimensional type A cluster combinatorics 2026
With Mikhail Gorsky
arXiv:2605.27263
4. Oriented matroids from type A cluster categories 2025
arXiv:2509.07883
3. Preorders on maximal chains: 2025
hyperplane arrangements, Cambrian lattices, and maximal green sequences
With Mikhail Gorsky
arXiv:2506.08858
2. Donaldson–Thomas invariants for the Bridgeland–Smith correspondence 2024
With Omar Kidwai
arXiv:2401.10093
1. A structural view of maximal green sequences 2023
With Mikhail Gorsky
arXiv:2301.08681

PRIZES AND SCHOLARSHIPS

- Gibbs prize for Mathematics & Philosophy** 2016
University of Oxford
For top performance in Mathematics papers
- Gibbs prize for Mathematics & Philosophy** 2015
University of Oxford
For top performance in Mathematics papers
- Academic scholarship** 2013
Worcester College, Oxford

INVITED CONFERENCE TALKS (IN PERSON)

- Higher Ginzburg DG algebras from dimer models and their symplectic incarnations** April 2026
Symplectic topology meets representation theory
University of Cambridge
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** May 2024
Algebraic Geometry Days in Glasgow
University of Glasgow
- Triangulations of cyclic polytopes through the lens of reconfiguration** August 2023
Minisymposium on Combinatorial Reconfiguration
International Congress on Industrial and Applied Mathematics
Waseda University, Tokyo
- A structural view of maximal green sequences** June 2023
Cluster structures in the North
University of Central Lancashire
- A structural approach to maximal green sequences** September 2022
Silting theory, algebras and representations
Charles University, Prague

INVITED CONFERENCE TALKS (VIRTUAL)

- Oriented matroids and maximal green sequences of type A cluster algebras** *January 2026*
Flash Talks in Representation Theory
- An algebraic interpretation of the higher Stasheff–Tamari orders** *November 2020*
International Conference on Representations of Algebras
Selected to give talk by scientific committee based on research snapshot

CONTRIBUTED CONFERENCE TALKS (IN PERSON)

- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *August 2024*
International Conference on Representations of Algebras
Shanghai Jiao Tong University
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *February 2024*
Clusters and their Applications in the North
University of Glasgow
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *January 2024*
Cluster Algebras and Its Applications
Mathematisches Forschungsinstitut Oberwolfach
- Triangulations of prisms and preprojective algebras of type A** *July 2023*
Combinatorial representation theory in Leeds
University of Leeds
- A structural view of maximal green sequences** *April 2023*
Maurice Auslander Distinguished Lectures and International Conference
Woods Hole, Massachusetts
- Cyclic polytopes and representation theory** *December 2022*
Representation theory of algebras session
Canadian Mathematical Society Winter Meeting, Toronto
- Algebraic interpretation of the higher Stasheff–Tamari orders** *July 2022*
Various issues on representation theory and related topics
Research Institute for Mathematical Sciences, Kyoto University
- Higher analogues of Grassmannian clusters** *April 2019*
Maurice Auslander Distinguished Lectures and International Conference
Woods Hole, Massachusetts
- Higher analogues of Grassmannian clusters** *April 2019*
New Connections and Directions in Representation Theory
University of Leeds
- Higher analogues of Grassmannian clusters** *April 2019*
Tropical Geometry meets Representation Theory
University of Leicester

CONTRIBUTED CONFERENCE TALKS (VIRTUAL)

- The first higher Stasheff–Tamari orders are quotients of the higher Bruhat orders** *January 2022*
Formal Power Series and Algebraic Combinatorics
Bar-Ilan University

SEMINAR TALKS (IN PERSON)

- Higher Ginzburg DG algebras from dimer models and their symplectic incarnations** *March 2026*
MAGIC seminar
Imperial College London
- Cluster algebras, cluster categories, and higher cluster categories** *January 2026*
Algebra and Representation Theory Seminar
University of Cambridge
- Mutation des carquois et groupes de tresses** *December 2025*
Séminaire de géométrie, représentations, algèbre, analyse, logique
Université de Caen Normandie
- Steenrod operations via higher Bruhat orders** *December 2024*
Algebra, Number Theory, Logic and Representation theory seminar
University of East Anglia
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *October 2024*
Algebraic Geometry Seminar
University of Cambridge
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *June 2024*
Geometric, Algebraic, and Physical structures around the moduli of Meromorphic Quadratic Differentials
Simons Center for Geometry and Physics
Stony Brook University
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *April 2024*
Leeds Algebra Seminar
University of Leeds
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *April 2024*
Algebra and Number Theory Seminar
University of Vienna
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *March 2024*
Paris Algebra Seminar
Institut de Mathématiques de Jussieu-Paris Rive Gauche
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *October 2023*
Sheffield Geometry and Physics seminar
University of Sheffield
- Donaldson–Thomas invariants for the Bridgeland–Smith correspondence** *October 2023*
EGRET seminar
University of Edinburgh
- Donaldson–Thomas theory of quivers with potential** *September 2023*
Exact Asymptotics Theoretical Sciences Visiting Program
Okinawa Institute of Science and Technology
- Higher-dimensional cluster combinatorics and representation theory** *August 2023*
OIST Representation Theory Seminar
Okinawa Institute of Science and Technology
- Combinatorics of KP solitons** *August 2023*
Exact Asymptotics Theoretical Sciences Visiting Program
Okinawa Institute of Science and Technology

A structural view of maximal green sequences*February 2023*

Algebra Group Seminar

University of Cologne

The higher Stasheff–Tamari orders on triangulations of cyclic polytopes *November 2022*

Séminaire du LaCIM

Université du Québec à Montréal

Cyclic polytopes and representation theory*November 2022*

Geometry and Mathematical Physics Seminar

University of Birmingham

Cyclic polytopes and representation theory*October 2022*

Pure Mathematics Seminar

Lancaster University

Mutation in higher cluster categories*October 2022*

Seminar@Aarhus

Aarhus University

Equivalence of maximal green sequences*April 2022*

Working Seminar in Algebra and Geometry

University of Cologne

Equivalence of maximal green sequences*February 2022*

Paris Algebra Seminar

Institut de Mathématiques de Jussieu-Paris Rive Gauche

An algebraic interpretation of the higher Stasheff–Tamari orders*November 2019*

Pure Mathematics Seminar

University of East Anglia

Higher analogues of Grassmannian clusters*April 2019*

Algebra and Number Theory Seminar

University of Graz

SEMINAR TALKS (VIRTUAL)**Oriented matroids and maximal green sequences
of type A cluster algebras***April 2026*

Seminario de Álgebra homológica e hierbas afines

University of Buenos Aires

**Oriented matroids and maximal green sequences
of type A cluster algebras***October 2025*

Combinatorics and Representation Theory seminar

Korea Institute for Advanced Study

Steenrod operations via higher Bruhat orders*May 2025*

Longitudinal Algebra & Geometry Open Online Seminar (LAGOON)

Oriented matroids from cluster-tilting objects in type A cluster categories *March 2025*

New Directions in Group Theory & Triangulated Categories

Donaldson–Thomas invariants for the barbell quiver with potential*June 2024*

Cologne algebra seminar

University of Cologne

Mutating cluster-tilting objects in $(d + 2)$ -angulated cluster categories*June 2022*

FD seminar

An algebraic interpretation of the higher Stasheff–Tamari orders Aachen–Bochum–Cologne Darstellungstheorie Seminar RWTH Aachen University, Ruhr-University Bochum, and University of Cologne	<i>January 2022</i>
Cyclic polytopes and higher Auslander–Reiten theory Tokyo–Nagoya algebra seminar University of Tokyo and Nagoya University	<i>December 2021</i>
The two higher Stasheff–Tamari orders are equal Working seminar in algebra and geometry University of Cologne	<i>May 2021</i>
The higher Stasheff–Tamari orders in representation theory Paris algebra seminar Institut de Mathématiques de Jussieu-Paris Rive Gauche	<i>April 2021</i>
New interpretations of the higher Stasheff–Tamari orders Diskrete Mathematik / Geometrie Forschungsseminar TU Berlin	<i>March 2021</i>

LECTURE SERIES (IN PERSON)

Introduction to Donaldson–Thomas theory for quivers with potential Lancaster University 12 lectures	<i>October 2023 – February 2024</i>
--	-------------------------------------

LECTURE SERIES (VIRTUAL)

Cyclic polytopes and higher Auslander–Reiten theory Tokyo–Nagoya algebra seminar University of Tokyo and Nagoya University 3 lectures	<i>June – July 2022</i>
---	-------------------------

POSTERS

From higher Bruhat orders to Steenrod cup-i coproducts <i>Formal Power Series and Algebraic Combinatorics</i> Ruhr-Universität Bochum	<i>July 2024</i>
Studying triangulations of even-dimensional cyclic polytopes via directed graphs <i>Formal Power Series and Algebraic Combinatorics</i> University of California, Davis	<i>July 2023</i>
An algebraic interpretation of the higher Stasheff–Tamari orders <i>Young Researchers in Mathematics</i> Online Winner of 1 st prize in poster competition	<i>June 2021</i>
An algebraic interpretation of the higher Stasheff–Tamari orders <i>Advances in Representation Theory of Algebras</i> Online	<i>May 2021</i>
New interpretations of the higher Stasheff–Tamari orders <i>Combinatorial algebra meets algebraic combinatorics</i> Université du Québec à Montréal Online	<i>January 2021</i>

An algebraic interpretation of the higher Stasheff–Tamari orders*November 2020**International Conference on Representations of Algebras*

“Research snapshot”

Online

New interpretations of the higher Stasheff–Tamari orders*September 2020**Virtual Heidelberg Laureate Forum*

Online

Representation theory and Grassmannians*March 2019**STEM for Britain*

Poster competition at UK Parliament

GRANTS AND AWARDS

London Mathematical Society Scheme 1

October 2023

Conference grants

Silting, simple-minded, and stability

Co-organised with Raquel Coelho Simões and David Pauksztello

GBP 4,000

Japan Society for the Promotion of Science

December 2021

Short Term Postdoctoral Research Fellowship

JPY 5,384,000

12 months awarded but 3 months taken

London Mathematical Society Scheme 8

April 2020

Postgraduate research conference grants

Categorifications in Representation Theory

Co-organised with Aran Tattar

GBP 2,500

ORGANISATION

*Simple-mindedness, silting, and stability**July 2024*

Co-organised with Raquel Coelho Simões and David Pauksztello

Ambleside Campus, University of Cumbria

*Köln Algebra and Representation Theory seminar**April 2021 – May 2022*

Co-organised with Daniel Schaefer

University of Cologne

Online

*Categorifications in Representation Theory**September 2020*

Co-organised with Aran Tattar

Online

*Mathematics PhD seminar**September 2019 – August 2020*

University of Leicester

*Interactions between Representation Theory**May 2019**and Homological Mirror Symmetry*

Local organising committee

University of Leicester

RESEARCH STAYS

Université de Caen Normandie
Visiting: Matthew Pressland
1 week

December 2025

Simons Center for Geometry and Physics
Stony Brook University
Geometric, Algebraic, and Physical structures around the moduli of Meromorphic Quadratic Differentials
Invited participant
2 weeks

June 2024

University of Copenhagen
Visiting: Guillaume Laplante-Anfossi
Host: Nathalie Wahl
1 week

January 2024

Okinawa Institute of Science and Technology
Exact Asymptotics: From Fluid Dynamics to Quantum Geometry
Theoretical Sciences Visiting Program
Invited participant
2 months

August – October 2023

Université du Québec à Montréal
Visiting: Hugh Thomas
2 weeks

November 2022

University of Aarhus
Visiting: Raphael Bennett-Tennenhaus and Amit Shah
1 week

October 2022

University of Cologne
Visiting: Sibylle Schroll and Aran Tattar
1 week

April 2021

Isaac Newton Institute for Mathematical Sciences
University of Cambridge
Cluster algebras and representation theory
Invited participant
1 week

December 2021

University of Graz
Visiting: Jordan McMahon
Host: Karin Baur
1 week

April 2019

REFEREEING

Algebra and Number Theory, Algebraic Geometry and Physics, Annals of Representation Theory, Communications in Algebra, Compositio Mathematica, Discrete Mathematics (x2), Electronic Journal of Combinatorics, Electronic Research Archive, Indagationes Mathematicae, Journal of Algebra (x2), Journal of Algebraic Combinatorics, Selecta Mathematica

JSPS Short Term Pre/Postdoctoral Fellowships (x9)

Quick opinions: Inventiones Mathematicae, Proceedings of the Edinburgh Mathematical Society, Selecta Mathematica

REVIEWING

AMS MathSciNet
15 reviews written

October 2022 – present

zbMATH
17 reviews written

September 2020 – present

TEACHING

Part II Number Fields
Supervisor
University of Cambridge

January 2025 – May 2025

Part III Commutative Algebra
Lecturer for 24-lecture course taken by ~50 students
Other duties: teaching problem sheet classes and setting examination
University of Cambridge

October 2025 – July 2026

Part IB Groups, Rings, and Modules
Lecture cover: 1 lecture
University of Cambridge

February 2025

Part II Algebraic Geometry
Supervisor
University of Cambridge

January 2025 – May 2025

Real analysis
Class teacher (online)
King's College London

September 2021 – December 2021

MA107: Quantitative methods
Class teacher
London School of Economics

September 2021 – December 2021

Calculus & analysis
Class teacher (online)
University of Leicester

October 2020 – December 2020

Second-year algebra
Class teacher
University of Leicester

January 2020 – April 2020

Calculus & analysis
Class teacher
University of Leicester

October 2019 – April 2020

Elements of number theory
Class teacher
University of Leicester

October 2019 – December 2019

Calculus & analysis
Class teacher
University of Leicester

January 2019 – March 2019

STUDENT SUPERVISION

Part III essay (Master's project)	<i>2025 – 2026</i>
University of Cambridge	
Two students	
Cambridge Summer Research in Mathematics	<i>2025</i>
Project: <i>Combinatorics of KP solitons</i>	
Students: Lilian Contius, Chris Yuzhen Yang	

COMMITTEE ROLES

College Affiliation Officer	<i>February 2026 – present</i>
Postdocs of Cambridge Society	
Mathematics & Philosophy Representative	<i>2015 – 2016</i>
Joint Consultative Committee with Undergraduates	
University of Oxford	
Undergraduate Representative	<i>2015 – 2016</i>
Joint Committee for Mathematics and Philosophy	
University of Oxford	
Undergraduate Representative	<i>2015 – 2016</i>
Mathematics Department Art Committee	
University of Oxford	

OUTREACH

<i>Target Oxbridge</i> lecture	<i>April 2026</i>
One-hour lecture giving a flavour of what Maths at Cambridge is like, delivered to high-achieving Year 12 students from Black heritage background	
University of Leicester Representative	<i>November 2018, 2019, 2020</i>
<i>Research in the UK afternoon</i>	
University of Cambridge	
Student Helper	<i>March 2013, 2014, 2015</i>
<i>Oxford and Cambridge Student Conferences</i>	

MEMBERSHIPS

Trinity College Postdoctoral Society	<i>January 2025 – present</i>
London Mathematical Society	<i>June 2019 – present</i>
Associate member	
3 reports on conferences written for the LMS newsletter	

OTHER

Interviewer for undergraduate mathematics admissions	<i>December 2025</i>
Two full days of interviews	
Emmanuel College, Cambridge	
CATAM examiner	<i>February – May 2025</i>
Marker for undergraduate computer projects	
University of Cambridge	

Programming	Python, Sage, Maple, node.js, Haskell Git L ^A T _E X
--------------------	---

Languages	(CEFR levels)
English	Native
French	B1
German	B1
Japanese	B1